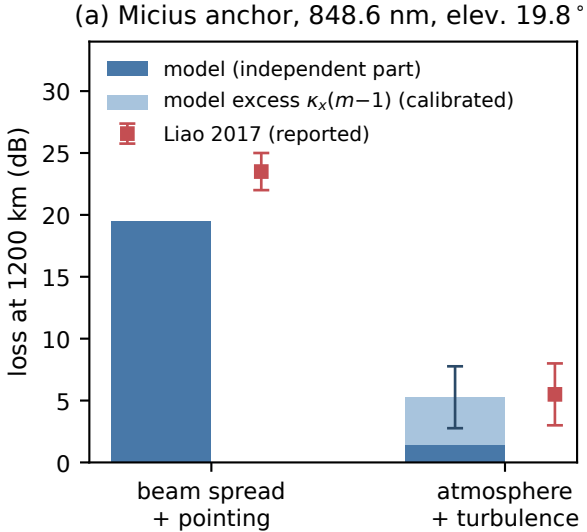
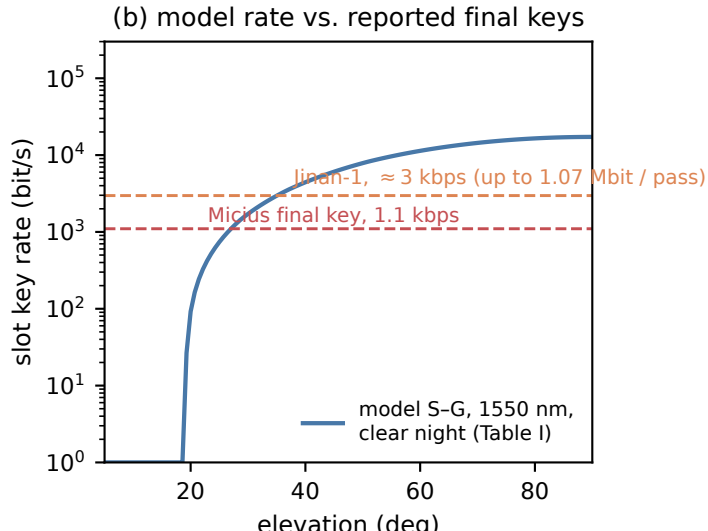
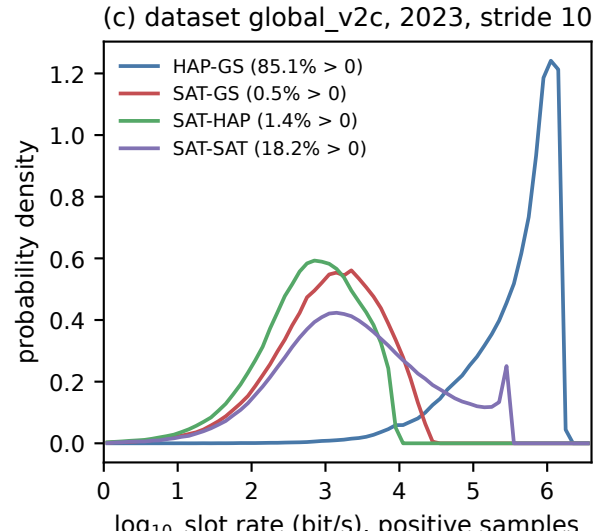

(a) Micius anchor, 848.6 nm, elev. 19.8°
model (independent part)
model excess $\kappa_x(m-1)$ (calibrated)
Liao 2017 (reported)
loss at 1200 km (dB)
0
5
10
15
20
25
30
beam spread
+ pointing
atmosphere
+ turbulence


(b) model rate vs. reported final keys
slot key rate (bit/s)
10^0
10^1
10^2
10^3
10^4
10^5
Jinan-1, ≈ 3 kbps (up to 1.07 Mbit / pass)
Micius final key, 1.1 kbps
model S–G, 1550 nm,
clear night (Table I)
20
40
60
80
elevation (deg)


(c) dataset global_v2c, 2023, stride 10
HAP-GS (85.1% > 0)
SAT-GS (0.5% > 0)
SAT-HAP (1.4% > 0)
SAT-SAT (18.2% > 0)
probability density
0.0
0.2
0.4
0.6
0.8
1.0
1.2
0
1
2
3
4
5
6
$\log_{10}$ slot rate (bit/s), positive samples